

Are Creativity and Schizotypy Products of a Right Hemisphere Bias?

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Published online February 14, 2002

Prior research has shown associations between higher creativity (e.g., semantic association, verbal fluency), higher schizotypy (e.g., magical ideation), and relatively stronger right hemisphere laterality measures—when each of the three pairings has been studied individually. Our prior study related creativity and schizotypy to signal detection theory response criterion aspects of laterality. The present study attempted to integrate findings regarding these three constructs and to replicate the signal detection theory laterality results by providing measures of all three constructs in a within subjects design. Participants were 60 undergraduates who completed a test battery including two measures of creativity, three measures of schizotypy, a lateralized lexical decision task, and a dichotic listening task. Results are consistent with individual differences in creativity and schizotypy being partly related to a response criterion favoring right hemisphere, possibly nonconscious, processing. Dichotic listening results revealed a strong association of better right hemisphere (left ear) localization ability and creativity. © 2002 Elsevier Science (USA)

Key Words: creativity; schizotypy; laterality; signal detection theory.

There is considerable evidence of an association among three variables in three quite different domains: creativity (and/or facilitation of distant semantic relations)—in the cognitive or intellectual domain, schizotypy—in the personality domain, and left versus right cerebral hemisphere laterality—in the neurophysiological domain. Theoretical accounts of these associations have posited that the common factor underlying individual variation in the three variables is a difference in semantic processing to favor distant or global associations, possibly by way of a loosening of response criterion to disinhibit weaker associations (Brugger & Graves, 1997b; Weinstein & Graves, 2001). According to this theory, increased availability of distant or less common semantic associations can result in both higher creativity scores on certain tests (e.g., of remote associates and verbal fluency) and also in higher scores on positive schizotypy tests (e.g., of magical ideation and unusual perceptual experiences). The underlying neurophysiological basis for this increased availability of distant semantic associations is, according to this theory, related to a pattern of cerebral laterality in which an individual has a relative weakening of left hemisphere dominance and strengthening of availability of right hemisphere processing. The nature of individual differences in hemispheric laterality could be either neuroanatomical, functional as in a signal detection theory criterion difference, or a combination.

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The theory of a common mechanism relating the three constructs would best be evaluated by studies in which multiple measures of all three constructs are obtained from the same individuals. However, almost all of the research in this area has looked at only two of the three constructs in each study and many studies have measured each construct with only one test. Many studies have reported an association between creativity and schizotypy (e.g., Brod, 1997; Duchene, Graves, & Brugger, 1998; Dykes & McGhie, 1976; Keefe & Magaro, 1980; Leonhard & Brugger, 1998; Miller & Chapman, 1983; Poreh, Whitman, & Ross, 1993–94; Schuldberg, French, Stone, & Heberle, 1988; Thalbourne & Delin, 1994; Tucker, Rothwell, Armstrong, & McConaghy, 1982). Other studies have reported an association between creativity and laterality (e.g., Brugger & Regard, 1995; Carlsson, Wendt, & Risberg, 2000; Crow, 1997; Harnad, 1972; Harpaz, 1990; Katz, 1983; Kravetz, Faust, & Edelma, 1998; Leonhard & Brugger, 1998; Martindale, Hines, Mitchell, & Covelio, 1984; Poreh, Whitman, & Ross, 1993–94). Still other studies have reported an association between schizotypy and laterality (e.g., Broks, 1984; Brugger, Gamma, Muri, Schafer, & Taylor, 1993; Brugger & Graves, 1997a; Falkai et al., 1995; Highley et al., 1998; Leonhard & Brugger, 1998; Mathew, Gruzeliier, & Liddle, 1993; Mattay et al., 1997; Persaud & Cutting, 1991; Richardson, Mason, & Claridge, 1997; Rossi et al., 1994; Rossi et al., 1992; Sakuma, Hoff, & DeLisi, 1996; Weisbrod, Maier, Harig, Himmesbach, & Spitzer, 1998). However, we know of only two previous studies that have measured all three constructs at once. Poreh, Whitman, and Ross (1993–1994) found that a group selected for high schizotypy had reduced left hemisphere (right ear) advantage on a consonant vowel dichotic listening test and also had higher scores on two of six creativity scales, but their creativity and laterality measures were not significantly associated. Weinstein and Graves (2001) provided two measures of each of the three constructs to unselected normal participants. They found an association between one schizotypy and one laterality measure and between one creativity and one laterality measure, but no significant association between their creativity and schizotypy measures. This study also used signal detection theory (SDT) to analyze the laterality data, apparently for the first time in research looking at the relation between laterality and creativity or schizotypy. The results indicated that the association between creativity on the Remote Associates Test (Mednick, 1959) and laterality (using a tachistoscopic lexical decision test) involved a difference in SDT *criterion*—normal individuals with higher creativity scores had lower criteria for decisions about stimuli presented to the right hemisphere (left visual field) than for stimuli presented to the left hemisphere (right visual field). This suggests that the mechanism for increasing the availability of right hemisphere processing at least partially involves lowering the criterion for accepting information coming from the right hemisphere. Most previous studies have used visual tachistoscopic methods to obtain a measure of laterality for comparison to creativity or schizotypy. A dichotic listening (DL) method was used by Poreh et al. (1993–1994), as discussed above. In addition, dichotic listening methods have been used to obtain measures of overall auditory function, rather than to obtain measures of lateralization of auditory language function. For example, schizophrenic persons had more errors on a DL task (Weigus & Harvey, 1988), and normal persons with higher paranormal beliefs (a schizotypy measure) had more overall DL errors (Skirda & Persinger, 1993; see also Roberts, 1993).

The purposes of the present study are to attempt to replicate results of the previous study (Weinstein & Graves, 2001) using a larger sample size, and to extend the investigation by including, for the first time in research on creativity and schizotypy, SDT analyses of both an auditory dichotic listening test and a visual lateralized lexical decision test.

METHOD

Participants

Participants were 62 University of Victoria undergraduate students receiving credit in Psychology 100. Participants were required to have a good understanding of the English language, no history of psychological/neurological disorder, and normal or corrected to normal vision and hearing. The first 62 candidates meeting the criteria were taken as test participants, two of whom were subsequently excluded (prior to data analysis) due to failure to follow instructions. Participants included in the final analysis were 21 men (3 of whom were non-right-handed) and 39 women (4 of whom were non-right-handed), with a mean age of 19.8. Handedness was assessed with the self-report Edinburgh Handedness Inventory (EHI), a scale of hand preference (Oldfield, 1971). A power analysis using the Gpower program (Erdfeiler, Faul, & Buchner, 1996) showed that a sample size of 60 provides a power of .78 to detect an *r* of .30 with an alpha level of .05 (one-tail).

Testing Instruments

Schizotypy. Schizotypy was measured with three self-report scales: Magical Ideation (MI) (Eckblad & Chapman, 1983), Perceptual Aberration (PA) (Chapman, Chapman, & Raulin, 1978), and Revised Social Anhedonia (SA) (Eckblad, Chapman, Chapman, & Mishlove, 1982). The MI items ask about paranormal and superstitious belief. The PA items relate to the individual's subjective experience of their body and sensory functioning. The SA items pertain to social attitudes and preferences. These three scales, along with 13 infrequency items (Chapman & Chapman, unpublished manuscript) were combined in a random format to form one questionnaire with 118 true or false items, as recommended by Chapman and Chapman (1983).

Creativity. Creativity was assessed with two written tests: a shortened version of the Remote Associates Test (RAT) (Mednick, 1959) that was devised for the previous study (Weinstein & Graves, 2001), and the Thurstone Written Fluency Test (TWFT) (Thurstone & Thurstone, 1962). The RAT presents the participants with three words and asks for a one word response that is associated with all three. In the original version there are 30 items and the test time is limited to 40 min. Due to time constraints, participants were tested with every third item (i.e., #1, #4, #7, etc.) for a total of 10 items and had 13 min (equal to approximately 1/3 of the original time of 40 min) to complete the test. The TWFT has two parts, termed the free and fixed conditions. The free condition asks the participant to write down, in a five-minute period, as many words as they can that begin with the letter S. The fixed condition asks for as many four-letter words beginning with the letter C as possible, allowing a four-minute period. Both conditions exclude proper nouns and variations of the same word.

Laterality. Laterality was assessed with two computer-generated recognition tasks, lexical decision, and dichotic listening. Stimuli for the lexical decision task were 15 four-letter English words and 45 four-letter nonsense words (Table 1). Stimuli were presented on a NEC 2A computer monitor, black on

TABLE 1
Lexical Decision Task Stimulus Pairs

Pair no.	Target trials		No target trials	
1.	soup	noik	snis	pand
2.	bike	smig	nawl	klum
3.	frog	crom	arum	fene
4.	nail	miln	bact	dace
5.	desk	zorg	tant	gleb
6.	girl	briv	chim	koes
7.	wood	noft	cile	dupt
8.	lamp	flad	vill	ress
9.	coin	rowt	koch	trem
10.	boat	noke	blig	jerd
11.	sofa	geat	wape	mert
12.	tree	swib	geez	bleg
13.	coat	losp	carf	quig
14.	note	boch	kets	erve
15.	fork	trib	wute	derp

Note. Each pair was presented twice, once as shown and once in the reverse order.

white background, for 150 ms. The computerized tachistoscope software was a modified version of programs described by Graves and Bradley (1988). View was binocular at a distance of 50 cm from the monitor. On each trial, two four-letter strings appeared, one extended 2.6 to 5.4° of visual angle to the left of fixation and one extended an equal distance to the right of fixation. On half the trials one string was a real word and one a nonsense word, for the other half of the trials both were nonsense words. All nonsense words were orthographically regular and pronounceable. Each word appeared once in the left visual field and once in the right. The 60 trials were randomized separately for each participant. Participants were asked to make judgments concerning whether or not a word appeared on each trial, with speed and accuracy considered of equal importance. Participants were required to make a lexical decision, that is, to press either the left or right keyboard shift key if a word was thought to have appeared on the left or right side or to press the spacebar if no word was thought to have appeared. The stimulus presentation and response requirements of this task were almost identical to those described in Leonhard and Brugger (1998) except that English nouns were used instead of German nouns. For the signal detection theory analysis, responses were scored as left visual field (LVF) hits or right visual field (RVF) hits for trials on which a real word appeared and there was a response on the correct side, or as LVF or RVF false positives if no word had been presented but a response was made on the left or right side. Trials on which a word was presented but the response was made on the wrong side were counted as misses.

The dichotic listening task (DLT) stimuli were six natural voice, consonant–vowel (CV) nonsense syllables produced by combining each of the six English stop consonants with the vowel /a/: /ba/, /pa/, /ga/, /ta/, /da/, and /ka/. The first author recorded each of these syllables with the Sound Blaster 16 board using the software of Graves and Allen (1996). This procedure allowed the CV syllables to be edited for equal length and onset of speech signal. The duration of each syllable was 300 ms. Recording of stimuli used a Shure SM58 microphone, Nakamichi MX-100 microphone amplifier, TTE 4500 Hz anti-aliasing filter and 20,000 Hz 16 bit sampling rate. There are 15 possible unique pairs (e.g., ba/ka) and 30 possible unique dichotic pairs (e.g., ba/ka, ka/ba). Each of these 30 dichotic pairs was presented three times, each time with a different visual pre-cue. The visual pre-cues consisted of the same six CV syllables presented either to match the syllable heard in the right ear, the syllable heard in the left ear, or to be different from both. This results in a total of 90 trials, presented in three blocks of 30. The dichotic pairings and frequency of each type of cue were equally dispersed among the three blocks, which were then randomly ordered within each block for presentation. The cue to dichotic sound stimulus onset asynchrony was 1 s and the intertrial interval was 3 s. Dichotic CV auditory stimuli were presented using the Sound Blaster 16 board with a specially modified version of the Graves and Allen (1996) software, which also generated the visual precue stimuli. The output of the Sound Blaster 16 board was connected to a headphone amplifier through TTE 4500 Hz low pass filters.

Visual stimuli were presented on an NEC 4FG monitor, black on white background, with a Pentium 166 computer. Sony MDR-V600 headphones were used to present the dichotic pairs. The average signal intensity for each channel was set to 80 dB sound pressure level (C weighting, Radio Shack 33-2050 sound level meter) as measured using a 1000 Hz calibration tone, and were balanced within 1 dB. The ambient noise level in the chamber was less than 50 dB. This dichotic task followed the procedure used by Hiscock, Inch, and Kinsbourne (1999a, 1999b), although different stimulus samples and equipment were used.

Participants were told that the visual pre-cue might or might not be the same as one of the two auditory stimuli directly following and were asked to decide if this was the case, and if so, to which ear the matching stimulus had been presented. Participants were asked to respond verbally, that is, to say “left” if they believed the matching stimulus had been presented to the left ear, “right” for the analogous case in the right ear, and “none” if they believed that neither of the auditory stimuli were the same as the visual pre-cue.

Procedure

Each participant was tested individually after supplying written informed consent. The scales and tests were administered in the following order: schizotypy questionnaire, DLT, RAT, TWFT, LDT, EHI. Participants were given specific instructions for each measure and allowed the opportunity to ask questions relating to test materials/procedures before each test began. Both the LDT and the DLT were administered after separate practice trials had been completed, to allow for familiarization with the stimuli and procedure of each. Including a short break between each test (approximately one minute), time for completion of the battery was approximately 70 min. After all tests were completed, participants were debriefed, allowed further questions, thanked for their time, and dismissed.

RESULTS

Replication Analysis

The first phase of data analysis, conducted to assess whether the current results replicated the authors’ previous findings, utilized data from all measures except the dichotic listening task, which was the only measure not used previously (Weinstein & Graves, 2001). Because of the high intercorrelation between MI and PA in this study ($r = .654, p < .001$) and in a previous large scale study (Chapman, Chapman, & Miller, 1982) and because of conventional practice in the field, MI and PA were combined (in our case by summing the Z scores of each test) to create the Per-Mag variable. SA was retained as a separate variable as no significant relationship was found between SA and the other two scales. The two conditions of the TWFT (free and fixed) also correlated significantly ($r = .382, p = .002$), and were collapsed to form a written fluency (WF) variable. The hit rate and false alarm scores obtained for the LDT were used to create the signal detection theory (SDT) laterality indices of A2 and B2, based on the A’ and B’’ signal detection theory measures of sensitivity and response criterion, respectively (Greir, 1971). A’ represents detection sensitivity, with higher values indicating better ability to discriminate presence or absence of target stimuli. A2, an indicator of lateral differences in detection sensitivity, was defined as R–L, where R = the arcsine transformation of A’ calculated for the right visual field (RVF) and L = the arcsine transformation of A’ calculated for the left visual field (LVF). Arcsine transformation was used to improve the normality of the calculated A’ measures (Hiscock, personal communication, April 2000). A2 indicates the relative recognition performance in each visual field, independent of guessing strategy, with lower values indicating a relatively weaker right than left visual field. B’’ represents the direction of the guessing strategy—values above 1 reveal a more stringent, conservative strategy, that is, a tendency to indicate no target detected. B2, an indicator of lateral differences in criterion, was defined as R–L, with R = B’’ for the RVF and L = B’’ for the LVF. B2 indicates the participants’ relative criterion for the two visual fields, with a higher value indicating a looser criterion for the LVF (right hemisphere) and stricter RVF criterion (left hemisphere). That is, higher B2 values indicate that the individual favors right hemisphere responses. Because right versus left difference scores would be meaningless if the overall accuracy rate is either too close to chance (50%) or too close to perfect performance (100%), only those participants whose LDT accuracy scores were above 60% and below 90% were retained. N was thus reduced to 46 for correlations involving the LDT indices. Table 2 gives the descriptive statistics for the raw data.

TABLE 2
Descriptive Statistics for the Raw Replication Data

Source	N	Minimum	Maximum	Mean	Std. deviation
MI	60	1	27	10.32	5.61
PA	60	0	20	6.62	4.50
SA	60	0	29	7.12	5.33
RAT	60	0	8	4.87	1.94
WF					
Free	60	30	80	48.92	9.99
Fixed	60	8	36	19.15	5.93
LDT laterality					
sensitivity	46	–.15	.46	.1633	.1562
criterion	46	–1.59	.55	–.4904	.4902

TABLE 3a
Correlation Matrix Showing Results for the Previous Study

	SA	RAT	WF	LDT A2	LDT B2
Per-Mag	.285	-.040	-.196	.186	.470**
SA		-.097	-.201	.103	.263
RAT			.643***	-.152	.420*
WF				.033	.157
LDT A2					-.178

* $p < .025$.

** $p < .01$.

*** $p < .001$; 1-tail.

The replication analysis was conducted on the same six variables analyzed in the original study: two for schizotypy (SA and Per-Mag), two for creativity (RAT and WF), and two for laterality (LDT A2 and B2). Data analysis followed the procedures of the previous study except we now used the arcsine transformation for A' and excluded LDT data outside the 60–90% accuracy range. In order to facilitate comparisons, the correlation matrix from the previous study (Weinstein & Graves, 2001) is reproduced as Table 3a. The correlation matrix for the present data is displayed in Table 3b. Comparison of the results of the two studies according to the Fisher z method described by Rosenthal and Rosnow (1984, pp. 494–495) shows that none of the correlations shown in Tables 3a and 3b differed significantly between the studies ($p > .1$) indicating that the present study replicated the previous results within statistical error limits.

Meta-analysis

A meta-analysis of the two studies was conducted to estimate the effect sizes (correlation coefficients) using the Fisher z method (Rosenthal & Rosnow, 1984, p. 498) weighting the results of each study by the degrees of freedom ($N-3$). Thus study 2 with twice the sample size is weighted more heavily. The significance level for the correlations was calculated using the method of weighted (by degrees of freedom) Z 's (Rosenthal, 1978). The results, displayed in Table 3c, reveal four statistically significant correlations, including two relating laterality criterion differences to both schizotypy and creativity.

Dichotic Listening Analysis

The DLT data were analyzed using SDT. In accord with recent studies by Hiscock et al. (1999a, 1999b), separate indices were defined for stimulus detection ability and

TABLE 3b
Correlation Matrix Showing Results for the Current Study

	SA	RAT	WF	LDT A2	LDT B2
Per-Mag	.054	.302**	.218*	-.118	.167
SA		.209	-.084	-.130	.071
RAT			.259*	-.154	.275*
WF				.160	.132
LDT A2					-.083

* $p < .05$.

** $p < .01$; 1-tail.

TABLE 3c
Meta-analysis of the Two Studies Showing Estimated Correlation Effect Size

	SA	RAT	WF	LDT A2	LDT B2
Per-Mag	.13	.196*	.086	−.010	.286*
SA		.112	−.122	−.063	.143
RAT			.401**	−.146	.330**
WF				.119	.141
LDT A2					−.118

* $p < .02$.

** $p < .005$; 1-tail.

stimulus location ability. The methods for defining hit and false alarm rates, and the SDT formulas, were as reported by Hiscock et al. (1999b, Appendix B), except that the square root transform of A' was not performed, since the formula in Hiscock et al. (1999b) was incorrect (Hiscock, personal communication, April 2000). Thus this DLT paradigm allows sensitivity measures ($\text{Arcsine}(A')$), and criterion measures (B''), in principle to be defined for both ears combined, for each ear alone, for the difference between ears, and, for all the foregoing, separately for detection-only and localization-only, which would be 16 distinct measures. There are, however, only six independent hit and false alarm rate variables, which means that a maximum of six derived variables can be independent of each other. Examination of the derived measure equations (Hiscock, et al., 1999b, Appendix B) shows that four independent detection-only measures are possible, plus two additional localization-only measures. As will be seen below, we examined four detection-only measures, retained three, plus two localization-only measures for a total of five, fewer than the maximum of six possible independent measures.

Consideration of the inability of the right minus left laterality formulas to yield meaningful differences if overall performance was constrained by floor or ceiling effects necessitated exclusion of the data of some participants. Detection and localization overall accuracy scores were calculated for each participant, with 50% indicating at-chance performance and 100% indicating perfect performance. The same inclusion criteria used for the LDT (accuracy $\geq 60\%$ and $\leq 90\%$) were applied to both localization and detection overall accuracy scores, resulting in $N = 59$ for detection and $N = 28$ for localization.

Syllable detection. The detection-only (disregarding localization) criterion and detection-only sensitivity laterality indices (right ear minus left ear) were found to be highly correlated, $r = -.83$, $p < .001$, indicating that detection-only sensitivity and detection-only criterion laterality could not be separated with this paradigm. Accordingly, we defined a single, simpler, measure of detection-only laterality as the right ear detection-only hit rate minus the left ear detection-only hit rate. This new variable, which presumably reflects both sensitivity and criterion effects, correlated $r = .95$ with detection-only sensitivity and $r = -.71$, $p < .001$ with detection-only criterion laterality indices, and showed a significant right ear advantage for detection-only hit rate, $t(58) = 5.1$, $p < .001$. For the 59 participants with useable DLT detection data, the one DLT detection-only laterality index (hit rate difference) did not correlate significantly ($p > .05$) with either creativity variable, or with either schizotypy variable, failing to accord with predictions. See Table 4 for results of all DLT variables.

Overall detection-only performance (combining both ears) showed that sensitivity and criterion, while significantly correlated ($r = .323$, $p = .012$ two-tail) only shared

TABLE 4
Correlation Matrix for Creativity, Schizotypy, and DLT Indices

	SA	Per-Mag	RAT	WF
Detection-only sensitivity (overall R + L) (<i>N</i> = 59)	.033	-.041	-.035	-.224
Detection-only criterion (overall R + L) (<i>N</i> = 59)	-.004	-.241*	-.016	-.051
Detection-only hit rate (laterality R - L) (<i>N</i> = 59)	.143	-.011	-.036	-.022
Localization-only sensitivity (overall R + L) (<i>N</i> = 28)	.004	-.069	.495**	.042
Localization-only sensitivity (laterality R - L) (<i>N</i> = 28)	-.014	-.114	.001	-.669***

* $p < .05$; 1-tail.

** $p < .01$; 2-tail.

*** $p < .001$; 2-tail.

10% variance, and hence these were kept as separate variables. While the theories under consideration appear to make no predictions for *DLT overall detection-only sensitivity*, this variable showed a significant correlation with written fluency. *DLT overall detection-only criterion* showed a significant correlation with Per-Mag, in the predicted direction.

Syllable localization. The large reduction in participants whose data was considered useable for DLT localization laterality indices reflects the difficulty of localization of stimuli in this task, as 31 participants (52%) were excluded because their overall localization accuracy scores were too close to chance (accuracy below 60%). Only one participant exceeded the maximum accuracy of 90%. The average localization accuracy in the present study (over all 60 participants) for both ears was 62%, which is similar to that found by Hiscock et al. (1999a) who observed approximately 65% accuracy on the same task.

A *DLT localization-only criterion laterality* index was not calculated since we could not conceptualize what such a quantity might represent in this two-choice paradigm. Consequently, only one localization-only laterality index was calculated. For the 28 participants with useable data, *DLT localization-only sensitivity laterality* correlated with the TWFT in the predicted direction ($r = -.669$, $p < .005$ one-tail). There was a significant right ear advantage for localization-only sensitivity ($t(27) = 3.2$, $p = .004$ two-tail). Thus, a lessening of the right ear advantage (i.e., relatively better left ear/right hemisphere performance) was associated with increasing written fluency performance. Overall (combining right and left ear data) localization-only criterion correlated ($r = .80$, $p < .001$, $N = 28$) with overall detection-only criterion (which reflects the fact that the relevant false alarm rates are identical for these two overall variables), hence no further analyses were performed for *DLT overall localization-only criterion*. Although no a priori predictions had been made for *DLT overall localization-only sensitivity*, a significant correlation was found with the RAT.

DISCUSSION

Replication Analyses

The schizotypy scale intercorrelations replicated our previous results (Weinstein & Graves, 2001) and were in accord with other results that show a closer relationship between Magical Ideation (MI) and Perceptual Aberration (PA) than for either of these two with Social Anhedonia (SA) (Cadenhead, Kumar, & Braff, 1996; Kwapil

et al., 1997). Bentall, Claridge, and Slade's (1989) factor analysis of schizotypal traits found that MI and PA loaded together on a factor reflecting positive psychotic symptomatology. In contrast, SA loaded on another factor reflecting social anxiety and cognitive disorganization, which were suggested to underlie negative forms of psychotic symptomatology. The schizophrenic versions of these two symptomatologies are thought to be related to different mechanisms of the illness (Costello, 1993). Magical Ideation relates to schizophrenic delusions and Perceptual Aberration relates to schizophrenic hallucinations, that is, both relate to the positive symptoms of acute schizophrenia. In contrast, Social Anhedonia relates to the chronic schizophrenic negative symptoms of withdrawal and lack of volition and affect. (See Venables & Rector, 2000 and Vollema & van den Bosch, 1995 for brief discussions of the history, theory, and factor structure of schizotypy.) MI and PA scores are highly correlated (Chapman, Chapman, & Miller, 1982) and (as in the current study) frequently combined to form a variable named "Per-Mag" (e.g., Cadenhead, Kumar, & Braff, 1996; Chapman & Chapman, 1987; Schuldberg, 1990; Stone & Schuldberg, 1990; Thalbourne, 1994). The creativity measures, the Remote Associates Test (RAT) and the Thurstone Written Fluency Test (TWFT), were significantly correlated, replicating the previous result. Nevertheless, these two tests may largely be measuring two distinct aspects of creativity since much of the variance is not shared. The low intercorrelations between the lexical decision and the dichotic listening laterality indices suggest that these are also not measuring the same construct. Earlier laterality research also reported low correlations between tachistoscopic visual field and auditory dichotic data (see Graves, 1983 for references and discussion).

The present study, like the previous study (Weinstein & Graves, 2001), tested predictions that the schizotypy, creativity, and laterality measures would be associated. Comparison of the results of the two studies (Tables 3a and 3b) according to the method of Rosenthal and Rosnow (1984) shows that none of the 12 correlations differed significantly between the studies (Table 3c). Thus, the results of the two studies are mutually consistent—neither study contains sufficient evidence to claim that it fails to replicate any result that was statistically significant in the other study.

A meta-analysis, combining the results of the two studies (Table 3c), showed that the first prediction, that creativity would be related to schizotypy, was supported. The Per-Mag schizotypy variable correlated with the RAT, while SA was not related to either creativity measure. This was the expected association, as it has been the "positive" schizotypal traits and not the "negative" traits that have previously been associated with creativity (Poreh et al., 1993–1994; Schuldberg, 1990; Schuldberg et al., 1988).

The second prediction was that schizotypy would be related to laterality. The previous study found that both Per-Mag and SA were associated with a looser criterion in the right hemisphere and a stricter left hemisphere criterion as measured by the LDT B2 criterion index (Weinstein & Graves, 2001), which accords with the theory of schizotypy as a product of increased reliance on right hemisphere semantic processing. These associations, while in the predicted directions, did not reach statistical significance in the present study. The meta-analysis, however, continues to show a significant association between higher schizotypy (Per-Mag) and criterion scores indicative of favoring right hemisphere processing.

The third prediction was that creativity would be associated with laterality, more specifically, that higher creativity would be related to increased right hemisphere ability (sensitivity) or increased reliance on the right hemisphere (criterion). This hypothesis was supported. The RAT with LDT criterion laterality correlation was statistically significant in the present study, replicating the significant correlation in the first study, and in the meta-analysis. This correlation implies that a higher ability

to relate dissimilar concepts on the RAT is related to a strategy of granting more confidence to information coming from the right hemisphere in lexical decision.

Dichotic Listening Analyses

The dichotic listening task scores were analyzed using signal detection theory, with separate analysis of syllable detection performance (with localization effects removed) and syllable localization performance (with detection effects removed). In view of the number of variables and correlations involved, we will present these results from a post-hoc perspective with the understanding that replication is required to confirm these findings. The present DLT data are similar to those of Hiscock et al. (1999a, 1999b) in showing significant right ear advantages for both detection and localization and also in the overall accuracy rates. Thus our implementation of the Hiscock paradigm appears to have produced comparable results.

The theoretical prediction was that laterality indices would show that individuals with relatively better (higher sensitivity) left ear/right hemisphere performance would have higher creativity and higher schizotypy scores. One of the four correlations testing this prediction was statistically significant. Localization-only sensitivity laterality correlated with the Thurstone Written Fluency Test in the predicted direction. This effect was quite large, accounting for 45% of the variance. Poreh et al. (1993–1994) also found a DL/creativity association, although with nonverbal creativity measures. A second theoretical prediction was that individuals with relatively looser criteria for left ear/right hemisphere information would have higher creativity and higher schizotypy. This prediction could not be tested because, first, the detection-only criterion laterality measure turned out to be confounded with sensitivity and, second, localization-only criterion laterality is an uninterpretable construct in the present paradigm. A third prediction was that individuals with a looser overall criterion (combining data from the two ears), for either detection or localization, would have higher creativity and schizotypy scores. One of the four correlations testing this prediction was statistically significant, overall detection-only criterion correlated with Per-Mag in the predicted direction. This effect likely does not represent pure detection, however, since this variable correlated highly with localization criterion within the subset of individuals for whom the localization data was useable. In summary, the DLT results support the theory in showing that relatively higher right hemisphere (left ear) localization sensitivity was associated with higher creativity (written fluency) and that an overall looser criterion was associated with higher schizotypy.

An intriguing nonpredicted result was also observed. Overall better localization-only sensitivity was associated with higher creativity on the RAT. Because both localization (Graves & Jones, 1992) and semantic priming (Greenwald, Draine, & Abrams, 1996) can occur in the absence of conscious awareness, better performance on both localization and creativity tasks may depend on better ability to utilize nonconscious (“intuitive”) information. Alternatively, because DL localization and the RAT were the two most challenging tasks in this study, another possible explanation is that better performance on both tasks may reflect better ability to maintain motivation or sustained attention (vigilance). However, since the mechanism for increased vigilance has itself been theorized to involve facilitation of nonconscious global awareness at the expense of conscious focal attention (Posner & Petersen, 1990) and to involve right hemisphere frontal areas (Posner & Rothbart, 1992), these two explanations may actually be different aspects of a single effect.

The results obtained from the dichotic listening task were likely constrained by the difficulty of the task, particularly for stimulus localization where 52% of the participants had to be excluded from analysis because of accuracy less than 10%

above chance. Nevertheless, two quite large localization effects were found with the data from the remaining 28 participants who could perform the localization task. This is only the second study ever conducted to attempt to relate dichotic listening performance to both schizotypy and laterality measures (the first was Poreh et al., 1993–1994), and the first to use signal detection theory for such comparisons. Future studies may be able to modify the DL paradigm to raise localization accuracy.

Response criterion, measured using signal detection theory, has turned out to be an important variable. The correlation between lateral difference in lexical decision task signal detection theory criterion and creativity (on the Remote Associates Test) was significant in the present study, replicating the significant result in the previous study. The new dichotic listening test was also successful in showing an association between lateral differences in response criterion with schizotypy (the Per-Mag measure). No studies prior to these have investigated response criterion in the context of creativity and schizotypy. It is also noteworthy that the *only* significant associations between the perceptual tests and schizotypy in our studies involved response criterion—both the visual lexical decision task and the auditory dichotic listening task revealed an association between decision criterion and schizotypy. Thus, the positive aspect of schizotypy was associated with response criterion on two distinct tasks in two different sensory modalities, suggesting that the schizotypy personality variable may be more related to functional (strategic) than structural (ability) individual differences. Because of these response criterion findings, and because creativity measures were associated both with sensitivity measures (dichotic listening) and with criterion measures (lexical decision), it seems clear that signal detection theory should be used to distinguish between stimulus sensitivity and response criterion aspects of perceptual performance in future studies of schizotypy and creativity.

The fact that auditory localization measures were found to be associated with both creativity measures suggests that localization should be included in future research. On the basis of the results from the two studies, Social Anhedonia (the negative/withdrawal aspect of schizotypy) and lexical decision laterality sensitivity appear not to be related to any of the other variables and these measures could be dispensed with in future studies. The inconsistency in finding statistically significant correlations across the two studies suggests that the effect sizes are relatively small (r approximately .2 to .4) and that sample sizes will need to be fairly large to detect these effects with sufficient power, or that meta-analytic techniques may be required. Finally, these results demonstrate that there is the need for further research that investigates all three constructs simultaneously. Only this approach can adequately test the theory of a common underlying mechanism leading to an individual pattern of functioning which includes being less lateralized, more creative, and higher in schizotypy.

The results of the present study, especially when combined in a meta-analysis with the results of our earlier study, are consistent with a theory that individuals with a lower overall criterion, particularly with a lower criterion for right hemisphere processing, may have higher positive schizotypy scores reflecting more belief in magical causation, may have more unusual perceptual experiences, and may have enhanced creativity in finding remote associations. Lowering of criterion may be adaptive in finding real links between concepts, but may also lead to Type II errors in false attributions of causation and hallucinatory perceptual experiences (cf., Brugger & Graves, 1997b). The new results showing an association between higher creativity and auditory localization may also reflect a concomitant loosening of dependence on conscious processing to facilitate utilization of nonconscious processing (cf., Graves & Jones, 1992). Thus, individual differences in schizotypy and creativity may reflect differences in dependence on conscious, focally attended, processing versus

accepting nonconscious, nonfocally attended processing. This cognitive difference may in part involve a functional reduction of dependence on left hemisphere dominant conscious systems.

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